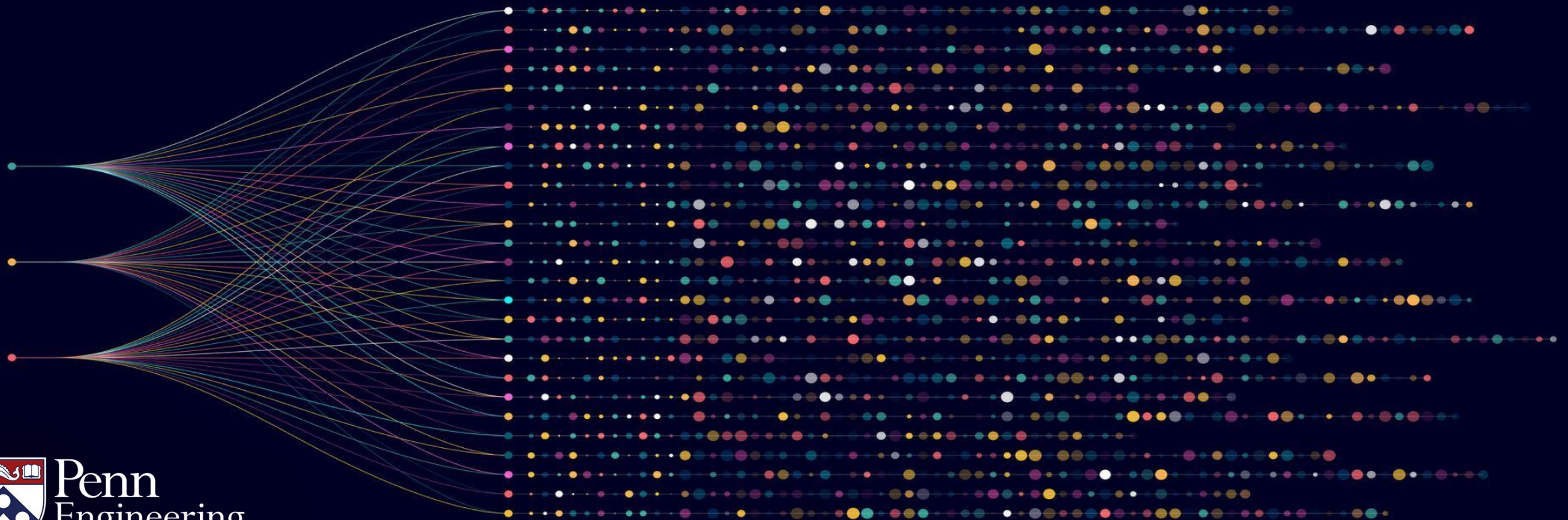


Algorithms for Big Data

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High Probability Guarantees



Objectives: Computing Large Averages

- Application of Chernoff/Hoeffding inequality

Chebyshev vs. Chernoff/Hoeffding

- Chebyshev Inequality: for random variable $\mathbf{Z}$ with $\mathbb{E}[\mathbf{Z}] = \mu$

$$\Pr[|\mathbf{Z} - \mu| \geq \epsilon] \leq \frac{\text{Var}[\mathbf{Z}]}{\epsilon^2}$$

- Chernoff/Hoeffding Inequality
 - If $\mathbf{Z}$ is a sum of q independent random variables between a and b :

$$\Pr[|\mathbf{Z} - \mu| \geq \epsilon] \leq 2e^{-2\epsilon^2 / (q(b-a)^2)}$$

Application of Chernoff/Hoeffding

- Chernoff/Hoeffding Inequality:

$$\Pr [|\mathbf{Z} - \mu| \geq \epsilon] \leq 2e^{-2\epsilon^2 / (q(b-a)^2)}$$

- Our setting: $\hat{\mathbf{S}} = \frac{1}{q} \sum_{t=1}^q X_{i_t}$ $b = 1/q$ $a = 0$

$$\Pr \left[\left| \hat{\mathbf{S}} - S \right| \geq \epsilon \right] \leq 2e^{-2\epsilon^2 / (q(1/q)^2)} = 2e^{-2\epsilon^2 q}$$

- Setting $q = \frac{\log(2/\delta)}{2\epsilon^2}$ gives desired bound!

Complete Description of Algorithm:

1. Sample $q = \frac{\log(2/\delta)}{2\epsilon^2}$ positions:

$$\mathbf{i}_1, \dots, \mathbf{i}_q \sim [n]$$

2. Output empirical average:

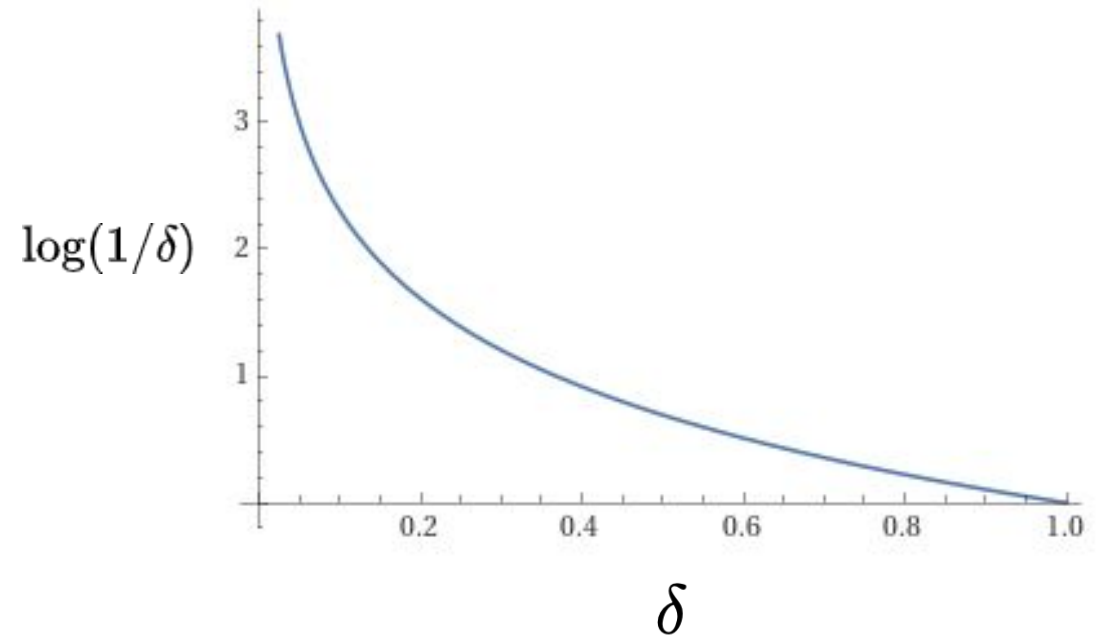
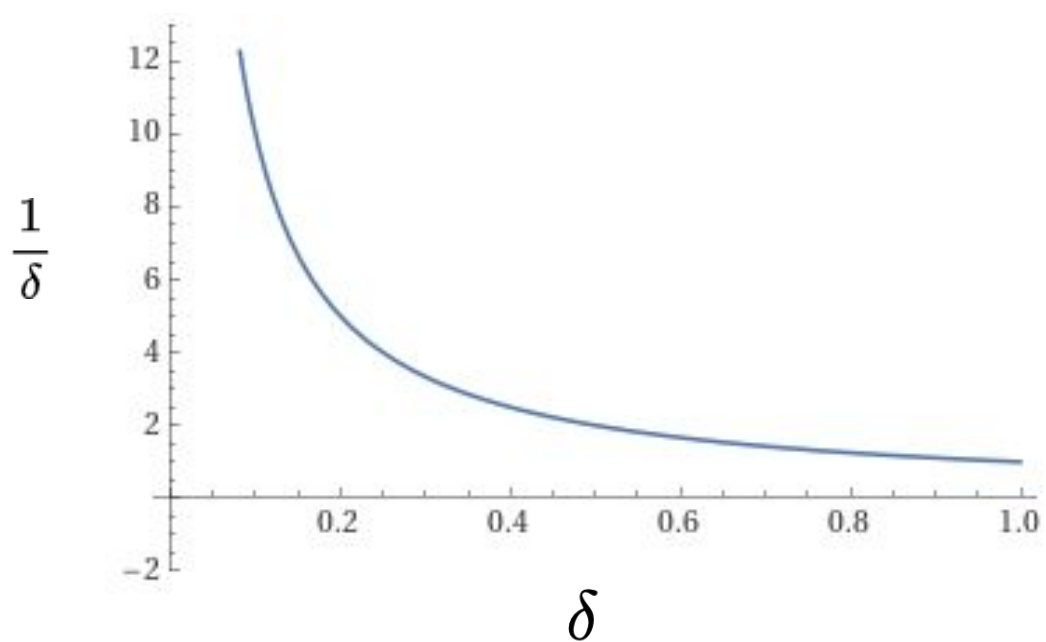
$$\hat{\mathbf{S}} = \frac{1}{q} \sum_{t=1}^q X_{\mathbf{i}_t}$$

- Asymptotically optimal dependence on accuracy and failure probability

$$O\left(\frac{1}{\delta\epsilon^2}\right) \text{ vs.}$$

$$O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$$

Inverse vs logarithmic dependence



- Logarithmic dependence is always better:
 - For failure probability 0.01 – inverse 100, log (base 2) is 4.6
- Much more pronounced when failure probability is really tiny

Take-aways:

- Two parameters: accuracy and failure probability

$$q = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$$

- Easy to make failure probability really *tiny*, but hard to make accuracy really *tiny*

		ϵ		
		0.1	0.01	0.001
δ	0.1	231	23K	2.3M
	0.01	460	46K	4.6M
	0.001	690	69K	6.9M

Why should you care?

- Relatively cheap to make randomized algorithm succeed with very high probability
- If algorithm fails with probability at most 2^{-300} , it will essentially never fail
 - with inverse relationship: intractable.
 - with logarithmic relationship: tractable.
- Very important when using repeated estimations